

VLM4RSDet: Collaborative Optimization with Vision-Language Model for Enhancing Remote Sensing Object Detection

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Abstract

Closed-set object detection in remote sensing imagery has made significant progress, but achieving high detection accuracy remains challenging. Vision-Language Models (VLMs), which possess rich prior knowledge, offer a promising solution to this challenge. However, most existing VLMs are designed for open-vocabulary tasks and exhibit inherent limitations when directly applied to closed-set scenarios, such as notable accuracy degradation and high deployment costs. To address these issues, we propose VLM4RSDet, a novel collaborative training framework that leverages vision-language model to enhance the performance of conventional closed-set remote sensing object detectors. Notably, during inference, VLM4RSDet only retains the standard object detection architecture, thus avoiding any additional deployment overhead. Furthermore, we introduce a Global-Local Cross-Attention (GLCA) module and a Learnable Hierarchical Prediction Strategy (LHPS) to further improve collaborative training performance. Extensive experiments on five benchmark datasets demonstrate the effectiveness and robustness of our approach. In particular, our method outperforms the state-of-the-art by 7.5% in $mAP_{0.5:0.95}$ on the VisDrone2019 dataset. Our code is available at <https://github.com/cszshi/VLM4RSDet>.

1. Introduction

With the rapid development of artificial intelligence and the increasing demand for spatial perception, remote sensing object detection has gradually emerged as one of the most critical areas in computer vision. Current research primarily focuses on addressing inherent challenges, such as suboptimal label assignment strategies [53, 58, 63], significant scale variations [3, 29], arbitrary object orientations [42], background noise interference [38], and so on. However, achieving high detection accuracy in closed-set remote sensing object detection remains a considerable challenge

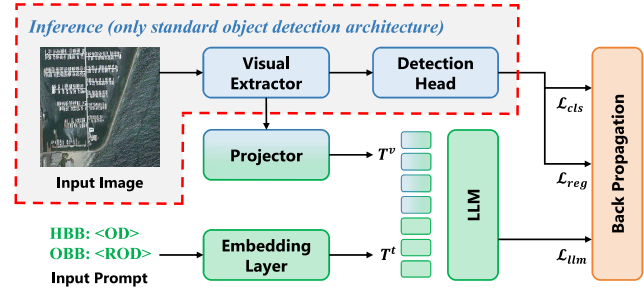


Figure 1. The overview of VLM4RSDet. In the training stage, we simultaneously optimize both the conventional object detection model and the vision-language model (VLM) to improve training effectiveness. In the inference stage, we only employ the standard object detection architecture (as shown in the red dashed area) to avoid additional deployment overhead.

that requires breakthrough solutions.

Recently, Vision-Language Models (VLMs) have garnered widespread attention from both academia and industry, owing to their rich prior knowledge and contextual reasoning capabilities. These models have demonstrated impressive performance across various tasks [1, 25, 44, 55]. In the field of remote sensing, a series of VLMs [21, 24, 34, 40, 57, 61] have been developed and successfully applied to tasks such as image captioning, object counting, and visual question answering. These advances highlight the potential of VLMs to enhance remote sensing object detection.

While several studies have indeed successfully applied vision-language models to object detection, their focus has predominantly remained on open-vocabulary scenarios, as exemplified by LLMDet [12], YOLO-World [9], Grounding DINO [36], and CoseDet [13]. Although these approaches share conceptual similarities with our collaborative optimization framework, they still exhibit a significant gap in both accuracy and efficiency compared to conventional models for closed-set remote sensing object detection. Specifically, the dependence on large language models or auxiliary modules increases inference costs. In addition, the detection accuracy of these methods still lags behind that of modern conventional closed-set detectors.

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Based on the above analysis, to break the accuracy bottleneck of conventional object detection models without increasing inference overhead, in this paper, we first propose a novel framework for collaborative optimization of vision-language model (VLM) and conventional object detection model in the field of closed-set remote sensing object detection, namely VLM4RSDet. Specifically, during training, the two models share a visual feature extraction network and back propagate the classification and bounding box regression losses of conventional object detection model, as well as the language modeling loss of VLM, to achieve collaborative optimization. In the inference stage, we only retain the standard object detection architecture to ensure that there is no additional overhead for model deployment. Additionally, in order to enhance the feature perception ability of VLM, we propose a Global-Local Cross Attention (GLCA) module. Finally, to address the issue of insufficient ability of VLM in predicting densely distributed objects in remote sensing images, we design a Learnable Hierarchical Prediction Strategy (LHPS) that dynamically assigns all objects within an image to various levels according to their sizes, thereby improving the output accuracy.

To the best of our knowledge, VLM4RSDet is the first collaborative training framework that leverages vision-language model to enhance conventional closed-set remote sensing object detectors, significantly improving detection accuracy without introducing additional inference or deployment overhead. Extensive experiments on five related datasets fully demonstrate the effectiveness and robustness of our method. We hope that VLM4RSDet will establish a strong baseline and inspire further research to explore the potential applications of vision-language models in conventional object detection tasks.

2. Related Work

2.1. Remote Sensing Object Detection

According to the annotation format of objects, remote sensing object detection can be divided into two categories: Horizontal Bounding Box (HBB) object detection and Oriented Bounding Box (OBB) object detection.

HBB Remote Sensing Object Detection. Remote sensing object detection often faces challenges such as extremely small object sizes, loss of effective information, and so on [8]. Recently, there are many improved methods have been proposed. For example, some of them focus on feature enhancement such as HANet [14], SR-TOD [5], and Bian et al. [2]. In addition, to address the problem of insufficient positive samples obtained by small objects, RFLA [53] models the ground truths and anchors through 2-D Gaussian distributions, and then measures the similarity through Wasserstein Distance (WD) or Kullback-Leibler Divergence (KLD). Based on the above method, KLDet

[63] and GA [58] further enhanced remote sensing object detection through more suitable label assignment strategies. To achieve better detection accuracy, the coarse-to-fine detection framework has been proposed, including UFPMP-Det [22], AdaZoom [54], YOLC [33], and so on. Due to these methods first process coarse detection for the entire image, and then process fine detection for these candidate regions, although they can significantly improve accuracy, the training and inference costs are also much higher.

OBB Remote Sensing Object Detection. Existing OBB remote sensing object detection methods can be roughly divided into two categories, one of which is the OBB object detectors formed by replacing certain components on the basis of HBB detectors, such as Rotated FCOS [47], Rotated RetinaNet [32], and Rotated Faster R-CNN [45]. Another category of methods involves modules specifically tailored to remote sensing object detection tasks. For example, ARC [42] proposed adaptive rotated convolution kernels to alleviate the phenomenon of arbitrary orientation of objects in remote sensing images, LSKNet [29] introduced large kernel convolutions to enhance the receptive field and contextual information, thereby improving the detection accuracy, PKINet [3] and LEGNet [38] from the perspective of model lightweighting, significantly reduced model parameters by optimizing the structure of backbone network.

Although existing methods have improved the performance of remote sensing object detection, significant bottleneck in accuracy still persists, particularly due to the lack of prior knowledge and other inherent limitations.

2.2. Vision-Language Model

VLMs for General Tasks. Recently, a series of vision-language models (VLMs) in general fields have been proposed, including CLIP [44], BLIP [25], Flamingo [1], GPT-4V [55], and so on. Early VLMs [25, 44] typically use the encoder-decoder architecture, while some recently proposed ones [1, 55] use decoder-only architecture, which consists of a visual encoder, a projector, and a large language model. Specifically, for general object detection tasks [46], existing methods leveraging VLMs can be broadly categorized into two paradigms: one involves fine-tuning to directly generate textual detection results, such as LMM-Det [26] and Florence-2 [51], while the other constructs open-vocabulary detectors by integrating language modules, such as LLMdet [12] and YOLO-World [9].

VLMs for Remote Sensing Tasks. With the advancement of general vision-language models, VLMs in the field of remote sensing have also gained growing attention in multiple tasks, including image captioning [21, 24], video captioning [57], visual question answering [40, 61], and so on. Although most VLMs for remote sensing are fine-tuned based on general VLMs, there are also some specifically designed methods according to the characteristics of remote sensing

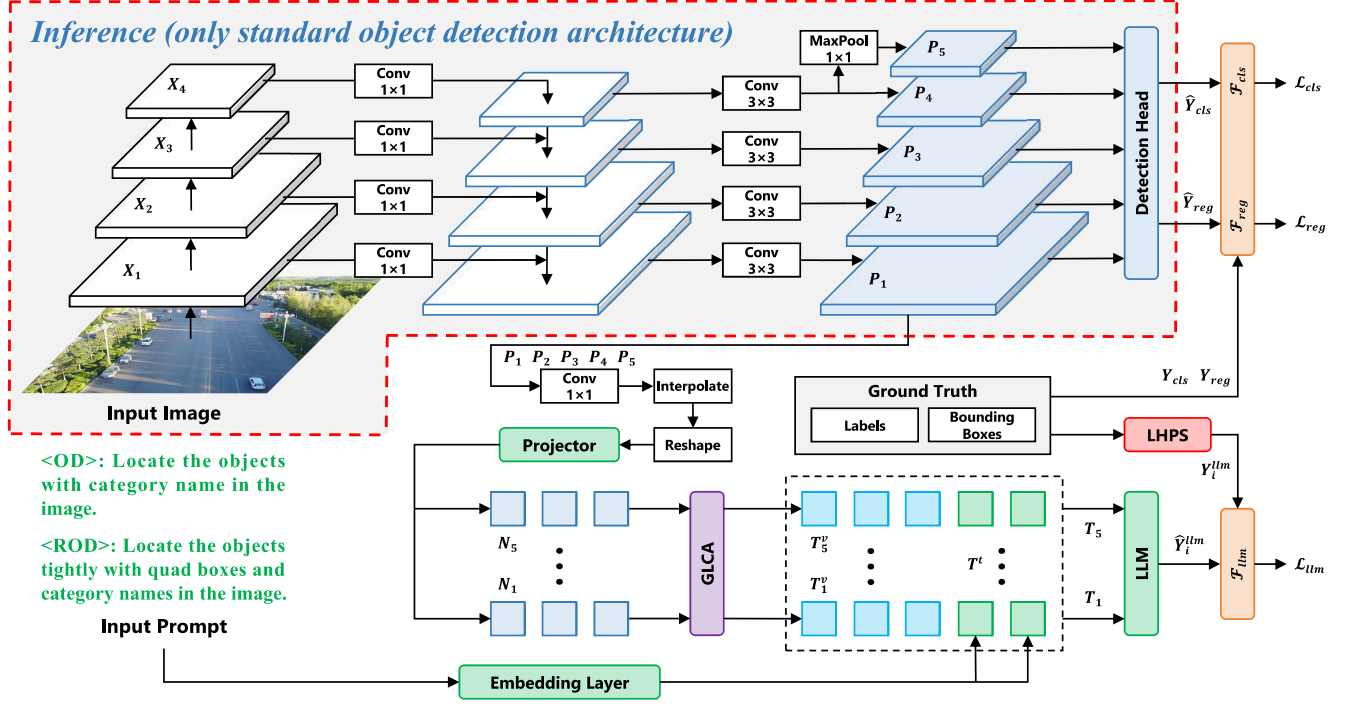


Figure 2. The overall structure of VLM4RSDet. In the training stage, the model is optimized with a composite loss function that integrates the conventional object detection losses ($\mathcal{L}_{cls}$ and $\mathcal{L}_{reg}$) with the language modeling loss $\mathcal{L}_{llm}$. This optimization is performed collaboratively, leveraging a series of shared multi-scale features P_i . In addition, we propose a Global-Local Cross Attention (GLCA, detailed in Section 3.2) and a Learnable Hierarchical Prediction Strategy (LHPS, detailed in Section 3.3) to enhance the feature perception ability and prediction output accuracy of vision-language model, respectively. In the inference stage, we only retain the standard object detection architecture (as shown in the red dashed area), which will not bring additional overhead to model deployment.

tasks. For example, EarthGPT [61] improved the visual encoder by simultaneously using ViT-based and CNN-based networks to extract features, thereby enhancing the perceptual ability of the model, VHM [41] constructed a specialized dataset to alleviate the hallucination problem. Currently, specialized VLMs designed for closed-set remote sensing object detection [27] remain relatively scarce. In addition, although some methods [23, 49] have achieved significant progress in accuracy of open-vocabulary remote sensing object detection tasks, their inference costs still lag behind most conventional detectors.

Therefore, a key research direction is to explore how to effectively leverage the rich prior knowledge of VLMs without increasing inference or deployment overhead.

3. Methodology

3.1. Model Architecture

The overall structure of our proposed collaborative optimization framework VLM4RSDet is shown in Fig. 2.

The Training Stage. For conventional object detection model, the input feature $I \in \mathbb{R}^{N \times 3 \times H_{in} \times W_{in}}$ (N is the batch size, the number of channels is 3, H_{in} and W_{in} repre-

sent the input height and width) is first fed into the backbone network. We denote the output features of each stage of the backbone as $X_i \in \mathbb{R}^{N \times C_i \times H_i \times W_i}$, taking ResNet-50 [17] as an example, $C_i = 64 \times 2^{i+1}$, $H_i = \frac{H_{in}}{2^{i+1}}$, $W_i = \frac{W_{in}}{2^{i+1}}$, $i = 1, 2, 3, 4$. Then, X_i is sent to the feature pyramid network (FPN) for further fusion. We denote the output features of FPN as $P_i \in \mathbb{R}^{N \times 256 \times H_i \times W_i}$, for standard FPN [31], $H_i = \frac{H_{in}}{2^{i+1}}$, $W_i = \frac{W_{in}}{2^{i+1}}$, $i = 1, 2, 3, 4, 5$. Finally, we feed P_i into the detection head to obtain the predicted output $\hat{Y}$ ($\hat{Y}_{cls}$, $\hat{Y}_{reg}$), and calculate the classification loss $\mathcal{L}_{cls}$ and bounding box regression loss $\mathcal{L}_{reg}$, as shown in (1).

$$\begin{aligned} \mathcal{L}_{cls} &= \mathcal{F}_{cls}(Y_{cls}, \hat{Y}_{cls}) \\ \mathcal{L}_{reg} &= \mathcal{F}_{reg}(Y_{reg}, \hat{Y}_{reg}) \end{aligned} \quad (1)$$

where Y_{cls} and Y_{reg} represent the labels and bounding boxes of each ground truth, $\hat{Y}_{cls}$ and $\hat{Y}_{reg}$ represent the classification and bounding box regression outputs, $\mathcal{F}_{cls}$ and $\mathcal{F}_{reg}$ represent the classification and bounding box regression loss functions.

For the projector and large language model (LLM), we use the corresponding modules from Florence-2-Base [51], which is a well pre-trained lightweight vision-language

model for various general tasks, especially object detection. In our proposed method, in order to promote collaborative optimization between conventional object detection model and vision-language model, we use each stage output of FPN P_i as visual inputs. To meet the input requirement of large language model, we first use 1×1 convolutions to convert the number of channels of P_i into d_v (d_v is 1024 for Florence-2-Base [51]), then use `F.interpolate` and `torch.reshape` sequentially to transform them into the shape of $\mathbb{R}^{N \times L^v \times d_v}$ (L^v equals to 32×32), finally, convert d_v to the input dimension of large language model d through the projector to obtain the processed visual features $N_i \in \mathbb{R}^{N \times L^v \times d}$. The above processes are shown in (2).

$$N_i = \text{Proj}(\text{Reshape}(\text{Inter}(\text{Conv}_{1 \times 1}(P_i)))) \quad (2)$$

where `Conv $_{1 \times 1}$` , `Inter`, `Reshape`, and `Proj` represent the 1×1 convolution, interpolate operation, reshape operation, and the projector, respectively.

In order to improve the feature perception ability, we feed N_i into the Global-Local Cross Attention module (Section 3.2) to obtain the enhanced features $T_i^v \in \mathbb{R}^{N \times L^v \times d}$. After that, we use `torch.concat` to concatenate the visual features T_i^v and the processed language feature $T^t \in \mathbb{R}^{N \times L^t \times d}$ (detailed in Section 3.3) along the second dimension to obtain the total input features $T_i \in \mathbb{R}^{N \times (L^v + L^t) \times d}$ of the large language model, as shown in (3).

$$T_i = \text{Concat}(T_i^v, T^t) \quad (3)$$

where `Concat` represents the concatenation operation.

After processing by the large language model, we denote the outputs as $\hat{Y}_i^{llm}$ and calculate the language modeling loss $\mathcal{L}_{llm}$, as shown in (4).

$$\mathcal{L}_{llm} = \sum_{i=1}^5 \mathcal{F}_{llm}(Y_i^{llm}, \hat{Y}_i^{llm}) \quad (4)$$

where Y_i^{llm} represents the target value obtained by the Learnable Hierarchical Prediction Strategy (Section 3.3), $\mathcal{F}_{llm}$ represents the language modeling loss function.

After obtaining the classification loss $\mathcal{L}_{cls}$, bounding box regression loss $\mathcal{L}_{reg}$, and the language modeling loss $\mathcal{L}_{llm}$, we can add them together and perform backpropagation, the total loss $\mathcal{L}$ is shown in (5).

$$\mathcal{L} = \mathcal{L}_{cls} + \mathcal{L}_{reg} + \alpha \times \mathcal{L}_{llm} \quad (5)$$

where α represents the weight used to balance the conventional object detection model and the vision-language model. Due to $\mathcal{L}_{llm}$ is much larger than $\mathcal{L}_{cls}$ and $\mathcal{L}_{reg}$, the value of α is chosen as 0.05 during the following experiments (detailed ablation study is shown in Section 4.3).

The Inference Stage. We remove the vision-language model and only retain the standard object detection architecture during inference. Therefore, compared with previous conventional object detection models, our method brings no additional inference or deployment overhead.

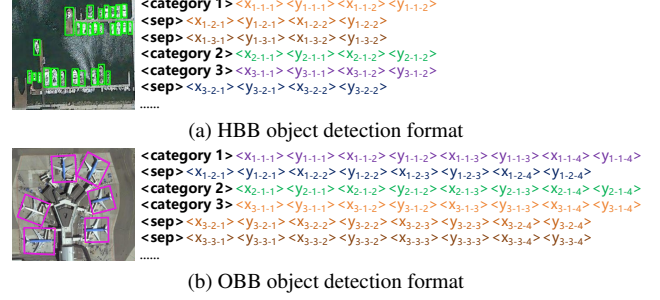


Figure 3. The target formats of the vision-language model. For HBB object detection, each bounding box consists of two coordinates, top left and bottom right. For OBB object detection, the coordinates of the four vertices are used in a clockwise direction. $\langle x_{i-j-k} \rangle \langle y_{i-j-k} \rangle$ represents the k -th coordinates of j -th object of i -th category.

3.2. Global-Local Cross Attention

To enhance the feature representation capability of vision-language model, we propose a Global-Local Cross Attention (GLCA) module, which can be integrated after the projector to refine the visual features. Specifically, in a hierarchy of features, highest-level one represents global information, while the current-level features retain corresponding local details. Therefore, we leverage the highest-level feature N_5 as the global context and dynamically fuse it with features from other levels via a cross attention mechanism. We employ the local feature N_i ($i = 1, 2, 3, 4$) as Query (Q), the global feature N_5 as Key (K) and Value (V), and fuse the global feature processed by attention mechanism with the local feature N_i of each stage, as shown in (6).

$$T_i^v = \begin{cases} \text{Softmax}\left(\frac{N_i N_5^T}{\sqrt{d}}\right) N_5 + N_i & \text{if } i = 1, 2, 3, 4 \\ N_5 & \text{if } i = 5 \end{cases} \quad (6)$$

where $T_i^v \in \mathbb{R}^{N \times L^v \times d}$ represents the enhanced visual feature, d represents the last dimension of N_5 (the dimension of input feature for large language model).

3.3. Learnable Hierarchical Prediction Strategy

In our proposed VLM4RSDet, we configure the language input of vision-language model (Florence-2-Base [51]) using the standard prompts `<OD>` and `<ROD>` for HBB and OBB object detection tasks, respectively. After processing by the embedding layer, we can obtain the language input feature $T^t \in \mathbb{R}^{N \times L^t \times d}$. For the output target value Y_i^{llm} of the vision-language model, we can construct them based on the labels and bounding boxes of each ground truth, the target formats are shown in Fig. 3. Specifically, we group the objects according to their categories. In cases where a category contains multiple objects, we employ the `<sep>` token to separate them.

The default Florence-2-Base [51] uses a single feature layer to predict all objects. In remote sensing images, due to the dense distribution of objects, an image often contains a large number of objects. In order to achieve more accurate prediction, we propose a Learnable Hierarchical Prediction Strategy (LHPS). Inspired by conventional object detectors, we divide all objects within an image into size-based groups, and use bottom-up features T_i ($i = 1, 2, 3, 4, 5$) as input of the vision-language model to predict each group. To improve adaptability, we introduce a set of learnable parameters β_i , which represent the proportion of objects in each group. Specifically, we first sort all objects in ascending order of size, and then determine the number of predicted objects M_i for each group based on the normalized scaling coefficient β_i^* , as shown in (7) and (8).

$$\beta_i^* = \frac{\beta_i}{\sum_{i=1}^5 \beta_i} \quad (7)$$

$$M_i = \lceil \beta_i^* \times M \rceil \quad (8)$$

where M represents the total number of objects within an image, $\lceil \cdot \rceil$ represents the round up operation to ensure that all objects can be included. Finally, according to the number of objects M_i in each group, we can obtain the target output Y_i^{llm} for each layer of the vision-language model.

4. Experiment

4.1. Experimental Setup

Datasets. For comprehensive comparisons, we use two HBB remote sensing object detection datasets (AI-TOD [48], VisDrone2019 [64]), two OBB remote sensing object detection datasets (DOTA-v1.0 [50], DOTA-v1.5 [50]), and one general object detection dataset (MS COCO 2017 [30]).

- **AI-TOD** [48] is a large-scale challenging benchmark for satellite remote sensing object detection, which includes 28,036 high-resolution images with 700,621 accurately annotated instances across 8 categories.
- **VisDrone2019** [64] is a widely used benchmark for unmanned aerial vehicle (UAV) remote sensing detection, which comprises 10,209 static images across 10 categories. It is a challenging dataset due to the fact that more than 60% of the objects are fewer than 20 pixels in size.
- **DOTA-v1.0** [50] contains 2,806 high-resolution remote sensing images with 188,282 annotated instances distributed across 15 categories, which contains 1,411, 458, and 937 images for training, validation, and testing.
- **DOTA-v1.5** [50] is an upgraded version of the DOTA-v1.0 dataset, retaining the same number of images but increasing the total annotations to 403,318 instances.
- **MS COCO 2017** [30] is a fundamental benchmark for general object detection tasks, containing 118,000 images and 860,000 instances across 80 categories.

Table 1. Main results on AI-TOD [48]. * means conducting 50 epochs of training.

Method	Publication	mAP _{0.5:0.95}	mAP _{0.5}	AP _{vt}	AP _t
<i>one-stage CNN-based:</i>					
RetinaNet [32]	ICCV 2017	8.7	22.3	2.4	8.9
FCOS [47]	ICCV 2019	12.6	30.4	2.3	12.2
<i>two-stage CNN-based:</i>					
Faster R-CNN [45]	TPAMI 2017	11.1	26.3	0.0	7.2
Cascade R-CNN [4]	CVPR 2018	13.8	30.8	0.0	10.6
DetectoRS [43]	CVPR 2021	14.8	32.8	0.0	10.8
<i>Transformer-based:</i>					
DINO [59]	Arxiv 2022	13.2	34.4	2.7	7.3
Crater-DETR [15]	TGRS 2024	18.1	42.1	-	-
Saliency DETR [20]	CVPR 2024	19.7	48.4	7.4	19.7
<i>VLM-based:</i>					
Florence-2-Base* [51]	CVPR 2024	22.3	38.4	-	-
Florence-2-Large* [51]	CVPR 2024	23.1	41.1	-	-
YOLO-World [9]	CVPR 2024	18.5	38.1	6.8	13.6
Grounding DINO [36]	ECCV 2024	19.6	45.3	12.4	17.6
LLMDet [12]	CVPR 2025	23.8	47.8	15.7	21.2
<i>state-of-the-art methods:</i>					
HANet [14]	TCSVT 2024	22.1	53.7	10.9	22.2
SR-TOD [5]	ECCV 2024	21.9	50.6	9.6	22.4
MENet [60]	TGRS 2024	23.2	56.2	9.7	23.9
TTFNet w/ GA [58]	TGRS 2024	24.2	53.9	10.4	25.2
DetectoRS w/ Bian et al. [2]	CVPR 2025	24.3	54.4	8.5	24.9
<i>ours:</i>					
RetinaNet w/ VLM4RSDet	-	14.9	36.5	9.1	16.4
FCOS w/ VLM4RSDet	-	21.4	49.9	8.1	24.5
Faster R-CNN w/ VLM4RSDet	-	23.7	55.7	16.3	27.5
DetectoRS w/ VLM4RSDet	-	28.5	59.9	18.8	31.6

Table 2. Main results on VisDrone2019 [64]. * means conducting 50 epochs of training.

Method	Publication	mAP _{0.5:0.95}	mAP _{0.5}	AP _{vt}	AP _t
<i>baseline:</i>					
RetinaNet [32]	ICCV 2017	7.9	15.9	0.1	0.7
FCOS [47]	ICCV 2019	14.1	25.5	0.1	2.1
Faster R-CNN [45]	TPAMI 2017	22.3	38.0	0.1	6.2
Cascade R-CNN [4]	CVPR 2018	22.5	38.5	0.5	6.8
DetectoRS [43]	CVPR 2021	25.7	41.7	0.5	7.6
<i>VLM-based:</i>					
Florence-2-Base* [51]	CVPR 2024	20.3	33.5	-	-
Florence-2-Large* [51]	CVPR 2024	21.8	35.2	-	-
YOLO-World [9]	CVPR 2024	14.3	23.6	2.8	7.1
Grounding DINO [36]	ECCV 2024	15.2	28.3	4.6	8.6
LLMDet [12]	CVPR 2025	20.4	34.1	7.4	12.8
<i>state-of-the-art methods:</i>					
RFLA [53]	ECCV 2022	27.4	45.3	4.5	12.9
KLDet [63]	TGRS 2024	28.5	50.7	-	-
Saliency DETR [20]	CVPR 2024	28.4	48.7	5.4	13.7
SR-TOD [5]	ECCV 2024	27.3	46.9	2.3	11.5
DN-FPN [35]	TGRS 2024	37.8	62.7	-	-
<i>ours:</i>					
RetinaNet w/ VLM4RSDet	-	23.5	41.0	2.7	6.8
FCOS w/ VLM4RSDet	-	22.9	40.7	3.6	9.2
Faster R-CNN w/ VLM4RSDet	-	26.7	47.9	8.3	14.8
DetectoRS w/ VLM4RSDet	-	31.4	53.0	9.3	16.5
DN-FPN w/ VLM4RSDet	-	45.3	70.5	10.7	18.6

Implementation Details. In this work, we choose the well pre-trained Florence-2-Base [51] as the vision-language model. Following previous research [27], we resize the visual input features to 32×32 with 1024 channels, and limit the maximum generation token length to 2048. During the collaborative training process, for AI-TOD and VisDrone2019 datasets, following previous researches [53, 63],

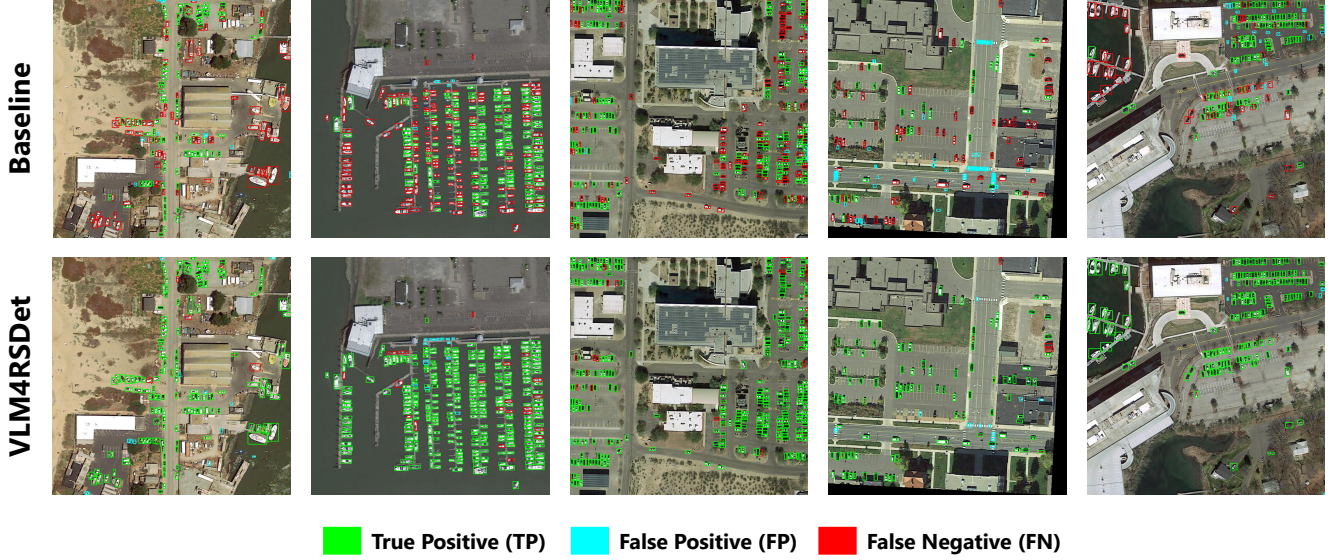


Figure 4. Visual comparisons on AI-TOD [48] dataset based on DetecToRS [43]. The green, blue, and red rectangles represent True Positive (TP), False Positive (FP), and False Negative (FN), respectively. An IoU threshold of 0.5 is used to determine a correct detection.

we employ the SGD with 0.9 momentum and 0.0001 weight decay as optimizer, the initial learning rate and batch size of each GPU are set to 0.02 and 2. The standard 2x and 1x training strategies are adopted for AI-TOD and VisDrone2019 datasets, respectively. For both DOTA-v1.0 and DOTA-v1.5 datasets, following the same data preprocessing and experimental settings in [3], we employ the AdamW with 0.05 weight decay as optimizer, the batch size of each GPU is set to 2. We conduct a 30-epoch training and the initial learning rate is set to 0.0002 with 0.1 times reductions at 16th, 22nd, and 27th epochs. For MS COCO 2017 dataset, the experimental settings are the same as those in VisDrone2019 dataset. For the above five datasets, we adopt the default partitioning method to split the training and testing sets. All of the following experiments are implemented using 4 NVIDIA RTX 4090 GPUs with the MMDetection [7] and MMRotate [62] toolboxes.

Evaluation Metrics. Following the evaluation metric established in previous research [53], we report the mean average precision at multiple IoU thresholds ($\text{mAP}_{0.5:0.95}$, $\text{mAP}_{0.5}$) for both AI-TOD and VisDrone2019 datasets. Additionally, we also provide the average precision for objects of varying sizes (AP_{vt} , AP_t) under the IoU range from 0.5 to 0.95. For DOTA-v1.0 and DOTA-v1.5 datasets, in accordance with [3], we report the mean average precision at 0.5 IoU threshold ($\text{mAP}_{0.5}$). For MS COCO 2017 dataset, we adopt the standard COCO evaluation metric [30].

4.2. Main Results

Performance on HBB Remote Sensing Object Detection Datasets. We conduct comprehensive experimental eval-

uation on a satellite HBB remote sensing object detection dataset AI-TOD [48] and an unmanned aerial vehicle (UAV) HBB remote sensing object detection dataset VisDrone2019 [64], as shown in Table 1 and Table 2, respectively. For AI-TOD dataset, we compare our method with conventional object detectors (CNN-based and Transformer-based), vision-language model (VLM)-based detectors, and some latest proposed state-of-the-art methods. Specifically, compared with conventional object detectors RetinaNet [32], FCOS [47], Faster R-CNN [45], and DetecToRS [43], our proposed VLM4RSDet improves detection accuracy $\text{mAP}_{0.5:0.95}$ by 6.2%, 8.8%, 12.6%, and 13.7%, respectively. Compared with the Florence-2-Large [51] that only uses vision-language model, our method can improve $\text{mAP}_{0.5:0.95}$ by 5.4% under fewer training epochs. For some latest proposed state-of-the-art methods, our VLM4RSDet can also achieve an average accuracy improvement of 5.4% in $\text{mAP}_{0.5:0.95}$. For VisDrone2019 dataset, we also conduct a series of experimental comparisons. It is worth noting that our VLM4RSDet outperforms the state-of-the-art method DN-FPN [35] by 7.5% in $\text{mAP}_{0.5:0.95}$ without increasing any inference overhead. Furthermore, we conduct qualitative comparisons on AI-TOD [48] dataset, as shown in Fig. 4. It is evident that our method significantly improves the detection accuracy, especially when dealing with complex scenes where objects are densely distributed.

Performance on OBB Remote Sensing Object Detection Datasets. The experimental comparisons on two classic OBB remote sensing object detection datasets, DOTA-v1.0 [50] and DOTA-v1.5 [50] are shown in Table 3 and Table 4, respectively. For DOTA-v1.0 dataset, compared

Table 3. Main results on DOTA-v1.0 [50]. All models are based on OBB object detectors. * means conducting 50 epochs of training.

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP _{0.5}
<i>one-stage CNN-based:</i>																
SASM [19]	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37	74.92
O-RepPoints [28]	87.02	83.17	54.13	71.16	80.18	78.40	87.28	90.90	85.97	86.25	59.90	70.49	73.53	72.27	58.97	75.97
Rotated FCOS [47]	88.52	77.54	47.06	63.78	80.42	80.50	87.34	90.39	77.83	84.13	55.45	65.84	66.02	72.77	49.17	72.45
<i>two-stage CNN-based:</i>																
ReDet [16]	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
Roi Trans [11]	89.01	77.48	51.64	72.07	74.4	77.55	87.76	90.81	79.71	85.27	58.36	64.11	76.50	71.99	54.06	74.05
Rotated Faster R-CNN [45]	89.40	81.81	47.28	67.44	73.96	73.12	85.03	90.90	85.15	84.90	56.60	64.77	64.70	70.28	62.22	73.17
O-RCNN [52]	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
<i>Transformer-based:</i>																
AO ² -DETR [10]	87.99	79.46	45.74	66.64	78.90	73.90	73.30	90.40	80.55	85.89	55.19	63.62	51.83	70.15	60.04	70.91
O ² -DETR [39]	86.01	75.92	46.02	66.65	79.70	79.93	89.17	90.44	81.19	76.00	56.91	62.45	64.22	65.80	58.96	72.15
ARS-DETR [56]	86.61	77.26	48.84	66.76	78.38	78.96	87.40	90.61	82.76	82.19	54.02	62.61	72.64	72.80	64.96	73.79
<i>VLM-based:</i>																
Florence-2-Base* [51]	61.21	55.16	31.75	46.19	54.61	51.49	50.87	62.58	55.87	59.63	38.62	44.25	35.92	47.87	41.58	49.17
Florence-2-Large* [51]	64.26	57.31	36.19	49.34	58.36	58.42	64.57	67.18	61.34	60.79	40.13	46.42	53.75	53.93	48.17	54.68
<i>state-of-the-art methods:</i>																
ARC-R50 [42]	89.40	82.48	55.33	73.88	79.37	84.05	88.06	90.90	86.44	84.83	63.63	70.32	74.29	71.91	65.43	77.35
LSKNet-S [29]	89.66	85.52	57.72	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82	77.49
DecoupleNet-D2 [37]	89.37	83.25	54.29	75.51	79.83	84.82	88.49	90.89	87.19	86.23	66.07	65.53	77.23	72.34	69.62	78.04
PKINet-S [3]	89.72	84.20	55.81	77.63	80.25	84.45	88.12	90.88	87.57	86.07	66.86	70.23	77.47	73.62	62.94	78.39
LEGNet-S [38]	89.87	86.32	56.36	78.67	81.34	85.29	88.53	90.89	87.31	86.46	70.38	68.48	78.34	82.00	70.25	80.03
<i>ours:</i>																
Rotated FCOS w/ VLM4RSDet	90.65	83.83	52.81	65.47	84.95	86.29	90.65	93.39	88.73	88.74	64.52	71.44	78.71	74.14	63.61	78.53
Rotated Faster R-CNN w/ VLM4RSDet	91.01	88.23	56.30	77.68	77.57	78.55	89.40	93.38	90.54	90.26	67.95	67.78	80.09	80.02	72.24	80.07
O-RCNN w/ VLM4RSDet	91.32	88.20	57.88	80.35	84.22	87.64	90.52	93.33	90.31	88.62	68.42	73.11	80.62	75.91	75.72	81.76
LEGNet-S w/ VLM4RSDet	94.14	91.03	58.81	82.58	86.54	90.35	93.56	96.27	93.13	91.27	70.13	75.16	82.79	77.82	77.46	84.07

Table 4. Main results on DOTA-v1.5 [50]. All models are based on OBB object detectors. * means conducting 50 epochs of training.

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP _{0.5}
<i>baseline:</i>																	
RetinaNet-O [32]	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
FR-O [45]	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
Mask R-CNN [18]	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42	62.67
HTC [6]	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40
AO ² -DETR [10]	79.55	78.14	42.41	61.23	55.34	74.50	79.57	90.64	74.76	77.58	53.56	66.91	58.56	73.11	69.64	24.71	66.26
ReDet [16]	79.20	82.81	51.92	71.41	52.38	75.73	80.92	90.83	75.81	68.64	49.29	72.03	73.36	70.55	63.33	11.53	66.86
<i>VLM-based:</i>																	
Florence-2-Base* [51]	41.13	48.51	31.68	42.79	29.91	44.28	46.35	51.88	45.35	39.47	35.53	42.15	44.30	43.05	31.96	24.13	40.15
Florence-2-Large* [51]	50.63	53.47	34.07	47.13	33.25	51.69	56.10	57.14	53.23	43.35	41.62	46.64	48.65	46.88	41.45	27.53	45.80
<i>state-of-the-art methods:</i>																	
LSKNet-S [29]	72.05	84.94	55.41	74.93	52.42	77.45	81.17	90.85	79.44	69.00	62.10	73.72	77.49	75.29	55.81	42.19	70.26
PKINet-S [3]	80.31	85.00	55.61	74.38	52.41	76.85	88.38	90.87	79.04	68.78	67.47	72.45	76.24	74.53	64.07	37.13	71.47
LEGNet-S [38]	80.48	85.04	55.64	74.86	52.64	82.16	89.11	90.88	84.55	68.91	66.21	74.16	77.44	74.68	65.76	43.75	72.89
<i>ours:</i>																	
RetinaNet-O w/ VLM4RSDet	69.28	81.65	53.29	72.16	50.45	74.47	85.13	87.29	76.05	66.41	59.68	70.91	74.38	72.36	53.79	40.78	68.01
FR-O w/ VLM4RSDet	76.51	80.67	50.73	74.24	49.92	77.26	84.45	86.47	75.48	65.89	61.40	68.15	73.08	68.75	68.63	43.62	69.08
LEGNet-S w/ VLM4RSDet	86.78	91.57	57.56	84.32	56.61	87.71	95.64	98.15	85.73	74.81	69.83	77.25	82.91	78.17	78.02	49.69	78.42

with one-stage detector Rotated FCOS [47] and two-stage detectors Rotated Faster R-CNN [45] and O-RCNN [52], our proposed VLM4RSDet can improve detection accuracy mAP_{0.5} by 6.08%, 6.90%, and 5.89%, respectively. Moreover, a significant performance gain is observed for our method, which surpasses the vision-language model (VLM)-based detectors Florence-2-Base [51] and Florence-2-Large [51] by 34.90% and 29.30% in mAP_{0.5}, respectively. Our VLM4RSDet achieves a new state-of-the-art mAP_{0.5} of 84.07%, outperforming LEGNet-S [38] and attaining the highest detection accuracy in multiple categories. For DOTA-v1.5 dataset, our method is compared

against a variety of detectors and achieves the best detection accuracy with a mAP_{0.5} of 78.42%.

Performance on General Object Detection Dataset. To further validate the robustness of our proposed VLM4RSDet, we also conduct experimental comparisons on a general object detection dataset MS COCO 2017 [30]. As shown in Table 5, it can be found that VLM4RSDet improves the detection accuracy mAP_{0.5:0.95} of baseline detectors RetinaNet [32], FCOS [47], and Faster R-CNN [45] by 4.8%, 5.1%, and 4.6%, respectively. The experimental results highlight that our proposed method is highly adaptive and robust across various object detection tasks.

Table 5. Main results on MS COCO 2017 [30].

Method	mAP _{0.5:0.95}	mAP _{0.5}	AP _s	AP _m	AP _l
<i>baseline:</i>					
RetinaNet [32]	36.5	55.4	20.4	40.3	48.1
FCOS [47]	36.6	56.0	21.0	40.6	47.0
Faster R-CNN [45]	37.4	58.1	21.2	41.0	48.1
<i>ours:</i>					
RetinaNet w/ VLM4RSDet	41.3	60.1	23.8	45.1	51.5
FCOS w/ VLM4RSDet	41.7	61.1	24.2	45.8	51.0
Faster R-CNN w/ VLM4RSDet	42.0	62.5	24.3	46.2	51.7

Table 6. Ablation study of each proposed module. All models are based on Detectors [43] and on VisDrone2019 [64].

New Structure	GLCA	LHPS	mAP _{0.5:0.95}	mAP _{0.5}	AP _{vt}	AP _t
			25.7	41.7	0.5	7.6
✓			28.5	46.5	5.8	9.3
✓	✓		29.8	48.3	6.2	11.4
✓		✓	30.2	49.8	8.0	13.6
✓	✓	✓	31.4	53.0	9.3	16.5

Table 7. Ablation study of different values of α . All models are based on Detectors [43] and on VisDrone2019 [64].

α	mAP _{0.5:0.95}	mAP _{0.5}	AP _{vt}	AP _t
0.03	28.1	48.3	8.2	14.8
0.04	29.8	51.4	8.9	15.7
0.05	31.4	53.0	9.3	16.5
0.06	30.3	51.7	8.8	15.6
0.07	29.6	50.4	8.5	15.0

4.3. Ablation Study

Effectiveness of Each Proposed Module. The ablation study of each module is shown in Table 6, New Structure refers to the collaborative optimization of conventional object detection model and vision-language model during the training stage. It can be found that the New Structure can improve the detection accuracy mAP_{0.5:0.95} by 2.8%. On this basis, GLCA and LHPS can further improve mAP_{0.5:0.95} by 1.3% and 1.7%, respectively. If all three modules are employed simultaneously, the detection accuracy can be significantly improved by 5.7%.

Performance of Different Values of α . In the training stage, we introduce a weighting parameter α to balance the losses from conventional object detection model and vision-language model, as defined in (5). To further determine the best value of α , we conduct an ablation study. From Table 7, it can be found that when α is set to 0.05, the model achieves the highest detection accuracy.

Analysis of Time and Space Cost During Inference and Training. As shown in Table 8, we employ three typical detectors (RetinaNet [32], FCOS [47], and Faster R-CNN [45]) as baseline methods, comparing their performance with and without VLM4RSDet, and additionally against Florence-2-Base [51], a fine-tuned vision-language

Table 8. Ablation study of time and space cost during inference and training based on VisDrone2019 [64]. The FLOPs and FPS (images per second) are counted under the input size equals to 1024×1024 . * means conducting 50 epochs of training.

Method	mAP _{0.5:0.95} ↑	Inference			Training
		Params ↓	FLOPs ↓	FPS ↑	FPS ↑
<i>baseline:</i>					
RetinaNet [32]	7.9	36.5M	212.9G	55.3	20.1
FCOS [47]	14.1	32.1M	201.6G	56.5	20.7
Faster R-CNN [45]	22.3	41.4M	239.0G	52.8	18.5
<i>VLM-based:</i>					
Florence-2-Base* [51]	20.3	140.2M	839.1G	45.2	17.3
<i>ours:</i>					
RetinaNet w/ VLM4RSDet	23.5	36.5M	212.9G	55.3	15.2
FCOS w/ VLM4RSDet	22.9	32.1M	201.6G	56.5	16.3
Faster R-CNN w/ VLM4RSDet	26.7	41.4M	239.0G	52.8	14.2

model for object detection. Notably, during inference, our VLM4RSDet introduces no additional overhead in parameters, FLOPs, or FPS compared with the baseline detectors, while its training time cost is kept within an acceptable margin. Compared with the fine-tuned vision-language model Florence-2-Base, VLM4RSDet achieves a substantial reduction in model parameters and FLOPs by averages of 73.8% and 74.0%, while boosting the detection accuracy mAP_{0.5:0.95} and inference FPS by averages of 4.1% and 21.4%. In addition, the lower training FPS of VLM4RSDet does not increase the total training time, as its required number of training epochs is far fewer than that of fine-tuning Florence-2-Base (12 epochs vs. 50 epochs).

5. Discussion and Conclusion

In this paper, we propose VLM4RSDet, a remote sensing object detection method that integrates the conventional object detectors with a vision-language model (VLM) through collaborative training. This approach addresses two key limitations: the low accuracy of conventional object detection models and the high deployment overhead of VLMs. Furthermore, we introduce a Global-Local Cross Attention module and a Learnable Hierarchical Prediction Strategy, tailored for enhancing vision-language model in detecting densely distributed objects within remote sensing images. Extensive experiments demonstrate that VLM4RSDet significantly improves detection accuracy while maintaining high efficiency, achieving state-of-the-art performance on multiple remote sensing object detection datasets.

Limitations and Future Work. Our proposed collaborative training method has effectively improved the accuracy of conventional object detectors but remains confined to closed-set tasks, as inference only uses the standard detection architecture. Future research will extend this framework by fully integrating vision-language models, aiming to achieve efficient open-vocabulary detection system.

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